

TABLE 1: CHARACTERISTICS OF THE FOUR DOMAINS

Characteristics	Domains			
	Customer (CR)	Functional (FR)	Physical (DP)	Process (PV)
Manufacturing	Attributes that customers desire	Functional requirements specified for the product	Physical variables which can satisfy the functional requirements	Process variables that can control design parameters
Materials	Desired performance	Required properties	Micro-structure	Process
Software	Attributes desired in the software	Output	Input variables and algorithm	Sub-routines
Organisation	Customer satisfaction	Functions of the organisation	Programs or office	People and other resources that can support the programs
Systems	Attributes desired of the overall system	Functional requirement of the system	Machines, components, or sub-components	Resources (human, financial, materials, etc)
Business	ROI	Business goals	Business structures	Human and financial resources

easier to control. We can represent this relationship as a design matrix ([DM]) and then map the properties as:

$$FR = [DM] \times DP \text{ and}$$

$$DP = [DM] \times PV$$

The independence rule is met in two ways based on the DM matrix:

1. Diagonal. This happens when there is a direct relationship between DP-FR or PV-DP. In this case, we can fulfil each FR (functional requirement) using an independent DP (design parameter), or a DP along with PV (process variable).

2. Triangular. The DM matrix being triangular allows us to define each relation independently by fol-

lowing a specific sequence.

Sometimes, you might have to change the order of elements in the DM (design matrix) to make a diagonal or triangular matrix. It's possible that the matrix has both triangular and diagonal parts. When that happens, you can divide the design matrix into sections and handle them independently.

When we organise the design matrix in the way mentioned above, it is called 'uncoupled.' However, if we can't arrange the relationship in a diagonal or triangular matrix, the design is called 'coupled.' Dealing with coupled designs is challeng-

ing. They can be really frustrating for user interfaces and, in the worst scenario, can result in products that are unusable.

Let me show you some simple examples of coupled

design to explain its issues. We all know about car controls. When the driver wants to go faster, he needs to send more fuel into the engine. But, to ensure the fuel burns completely, the fuel injection time and ignition also have to be adjusted simultaneously.

Picture how challenging driving could be if, whenever the driver wanted to change the speed, they had to fiddle with two separate controls for fuel flow and ignition timing. Thankfully, in older cars, this double control was managed by the carburettor and distributor. Nowadays, modern cars have an onboard computer that handles the fuel and ignition automatically, so the driver doesn't need to worry about those specific controls. All they have to do is press the accelerator to speed up or slow down the car. It's much simpler and easier for the driver!

In the old days, wash basins and showers had two separate controls for hot and cold water. This meant that when you adjusted the water flow, the temperature would also change. Nowadays, things are much easier with modern plumbing. We use combined taps that have a single lever. To adjust the water temperature, you simply turn the lever, and to change the water flow, you lift or lower the lever. These improvements were made possible by separating the controls,

$$\begin{bmatrix} FR_1 \\ FR_2 \\ \vdots \\ FR_m \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \\ \vdots \\ DP_n \end{bmatrix}$$

Fig. 2: Design matrix

$$\begin{bmatrix} FR_1 \\ FR_2 \\ FR_3 \\ FR_4 \\ FR_5 \\ FR_6 \\ FR_7 \end{bmatrix} = \begin{bmatrix} X & & & & & & \\ X & X & & & & & \\ X & X & X & & & & \\ X & X & X & X & & & \\ & & & & X & & \\ & & & & & X & \\ & & & & & & X \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \\ DP_3 \\ DP_4 \\ DP_5 \\ DP_6 \\ DP_7 \\ DP_8 \end{bmatrix}$$

Fig. 3: Uncoupled design matrix